FoveaWAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

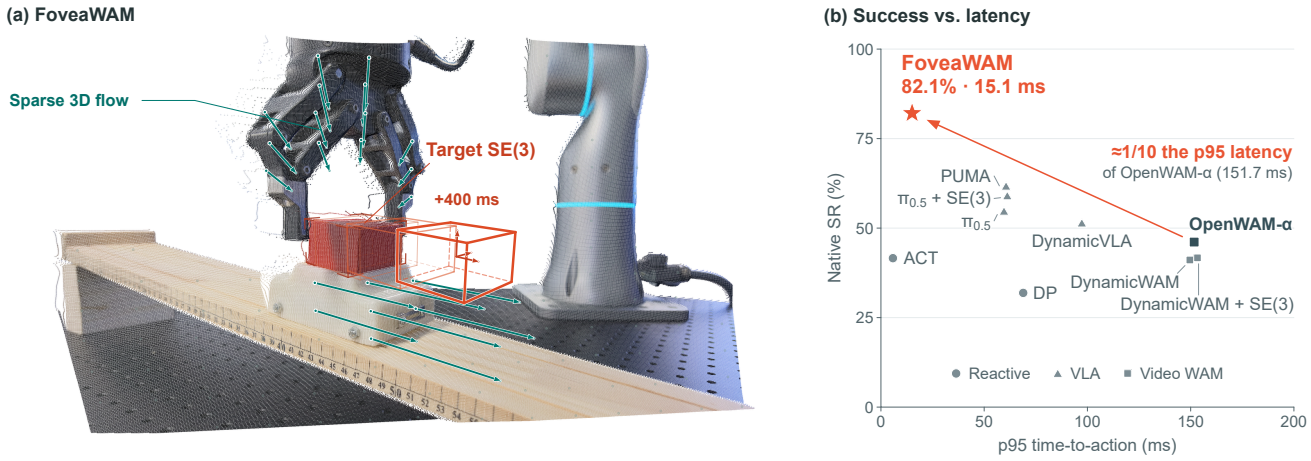


Figure 1. **FoveaWAM allocates prediction precision by task role.** (a) A real-robot scene reconstructed from RGB with MoGe-2 [44], with illustrative target SE(3) futures and sparse 3D flow. (b) Native SR versus p95 time-to-action from Table 1.

Abstract

To manipulate a moving object, a policy must predict the scene at the moment its action executes, yet every millisecond spent predicting pushes that moment further away. Existing policies either forecast the whole scene at uniform fidelity and act late, or act fast by ignoring everything but the target. Our key insight is that what a policy must predict decouples by role: only the target's pose, which decides contact, needs millimeter precision, while the surroundings matter only through the clearance they leave and the motion they impart. We present FoveaWAM, a native 4D world–action model that allocates precision by role: it models the target's past and future motion as a metric SE(3) trajectory and the surroundings as sparse 3D flow, and relies on the current frame for everything static. This compact representation lets FoveaWAM act in 15.1 ms on one RTX 5090, about $10\times$ *faster than video world–action baselines, giving the scene little time to move. To evaluate each policy under its own latency, we introduce OnTimeBench, a dynamic manipulation benchmark in which the scene keeps moving while the policy computes. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, and also leads on DOMINO and real-robot tasks.*

1. Introduction

To manipulate a moving object, a robot must act on a scene it has not yet seen. Objects keep moving while the policy perceives, computes, and acts, so every action lands in a later scene than the one observed. A ball rolling down a ramp at about 0.9 m/s moves 18 cm during a 200 ms inference; contact tolerates millimeters. The policy must therefore predict where the scene will be when its action executes. But prediction is itself computation, and its cost lengthens the interval it must cover: the longer a policy computes to anticipate motion, the further ahead it must anticipate. Existing dynamic policies settle this tension at one of two ends (Fig. 2). Some forecast the whole scene at uniform fidelity, as future video, latents, or 3D dynamics [29, 30, 45, 50, 55, 58]; they capture everything but act late. Others forecast only the manipulated object [14, 17]; they act sooner but miss the surroundings that block or redirect it. Either way, every part of the scene that is predicted costs the same, so knowing more means acting later. We ask instead: *what must a policy know about a moving scene, and how precisely, to make contact in time?*

Our key insight is that the answer decouples by role (Fig. 1). The target decides contact, so its pose at the moment of execution must be known to millimeter precision. The surroundings matter only through the clearance they leave

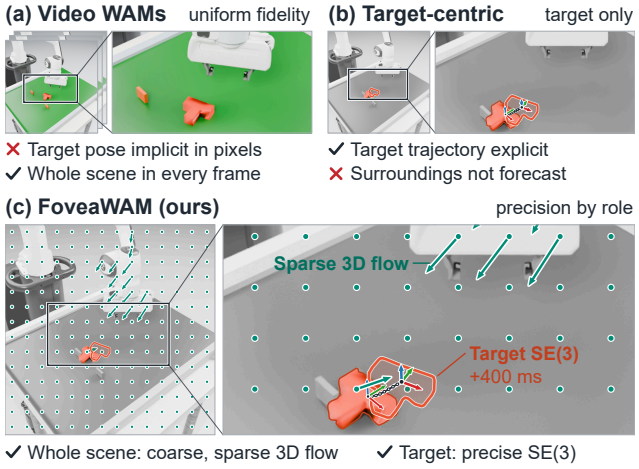


Figure 2. **Where forecast precision goes.** Three forecast representations in one recorded OnTimeBench scene. (a) Video WAMs leave the target pose implicit in future frames or latents [45, 50]. (b) Target-centric predictors forecast only the target's trajectory [14, 17]. (c) FoveaWAM forecasts sparse 3D scene flow (teal; dots: displacement $\leq$ 2 mm) and a precise SE(3) target trajectory (orange; axes at 0 and 400 ms). Only one flow sample lands on the target. Overlays come from the recording. Both forecasts are used only during training.

and the motion they impart to the target, so their motion need only be known coarsely. Everything static is already in the current frame and needs no prediction at all. What must be known precisely is therefore one object's trajectory, not the scene, and a policy that spends its precision there can be exact where contact is decided and still act in time. Biological vision makes the same allocation: the fovea resolves fine detail only where gaze lands, the periphery stays coarse yet alert to motion, and gaze runs ahead of the hand to where contact will occur [12, 19, 23, 33, 38].

We build this allocation into **FoveaWAM**, a native 4D world–action model whose representation follows the three roles (Fig. 2c). The target is described by a metric SE(3) trajectory: a short pose history supplies its past motion in place of past image frames, and the model learns to forecast its continuation. The surroundings are described by sparse 3D flow, and everything static by the current frame alone. Both forecasts supervise the representation during training and are dropped at inference [54]. As a result, FoveaWAM acts in 15.1 ms on one RTX 5090, about 10$\times$ faster than video world–action models, leaving the scene little time to move.

Measuring this advantage requires an evaluation in which latency has a cost. The default evaluators of DOMINO and PhysMani-Bench pause the world while the policy computes [14, 55], so a slow policy is scored as if it were instant. We introduce **OnTimeBench**, 14 dynamic manipulation tasks in which the world never pauses: the scene keeps moving for as long as each policy computes, and every method is evaluated at its native latency. The change of protocol reorders existing policies: DynamicWAM with SE(3) history beats the lightweight ACT [60] by 23.4 points with the world paused, but by only 0.1 points at native latency. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline. Its lead holds with the world paused and on DOMINO [14], and it succeeds in 37.5% of trials across six real-robot tasks. Ablations support the allocation: coarsening the target's forecast or removing its pose history lowers success. Uniform fine flow recovers overall success but trails on both the rotation-sensitive and context-active task groups.

Our contributions are threefold:

- We show that what a dynamic policy must know decouples by role: millimeter precision for the target's pose at execution time, coarse motion for the surroundings, and no prediction for the static scene.
- We present **FoveaWAM**, a native 4D world–action model that allocates precision by role and acts in 15.1 ms, about 10$\times$ faster than video world–action models.
- We introduce **OnTimeBench**, a dynamic manipulation benchmark that evaluates each policy at its native latency and reorders existing policies relative to paused evaluation. FoveaWAM reaches 82.1% success on it, 20.7 points above the strongest baseline, and also leads on DOMINO.

2. Related Work

Dynamic manipulation. Acting on moving objects, long handled by engineered tracking and interception [1–3, 32, 36], is now learned from data [9, 25, 48]. Learned policies take motion from image-space cues, such as optical flow in PUMA [14] and DynamicWAM [29], or act on test-time forecasts of the scene [30, 40, 55, 58]. FoveaWAM instead reads the target's metric SE(3) pose history and observes only the current frame. Execution schemes [6, 27, 35, 41, 48] change when actions are applied, not what the policy represents, and are complementary. DOMINO [14] and PhysMani-Bench [55] provide diverse dynamic tasks but by default pause the simulation until the policy returns; OnTimeBench advances it during inference, so each policy is evaluated at its own latency.

Forecasting in world–action models. WAMs generate future video and actions in one network [24, 45, 50, 62], building on video prediction as a policy objective [42, 47]. Because video is costly to generate, recent work forecasts cheaper targets, such as features, flow, object slots, or compact latents [28, 39, 51, 53, 57, 61], and some use the forecast only in training [14, 54, 59]. In all of them the forecast lives in pixel, feature, or latent space, and the target's future pose stays implicit. FoveaWAM keeps the training-only use but

forecasts a metric 4D state whose precision follows role, so the target's future pose is supervised explicitly.

Object-centric and geometric motion representations. Metric 3D inputs improve precision and sample efficiency [34, 56], and motion has been represented by point tracks [4, 46, 49], 3D flow [15, 18, 52, 55], object keypoints or SE(3) trajectories [17, 31, 37], and rigid-body decompositions [7, 8]. Each applies one representation to the whole scene, or models the target alone [14, 17]. DOCIR separates robot, target, and obstacles, but only as input encodings [13], and foveated robot perception allocates image resolution around a gaze point [11, 20, 22]. FoveaWAM allocates the precision of what is predicted: coarse sparse flow for the scene and a metric SE(3) trajectory for the target.

3. FoveaWAM

We formulate dynamic manipulation as generating time-indexed command chunks (Sec. 3.1) from a representation supervised, at the execution times of these commands, with the scene's sparse 3D flow and the target's SE(3) trajectory (Fig. 3). We describe the metric inputs (Sec. 3.2), this supervision (Sec. 3.3), and how an action stream reads the representation to generate the chunk (Sec. 3.4).

3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query t, a policy π learned from demonstrations reads an RGB image I_t and a depth map D_t captured at time τ_t, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of H commands

$$(\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command j is meant for time $\tau_t + j\Delta_a$, with control period Δ_a.

Each task specifies a *target*, whose pose the robot must match within a contact or containment tolerance. The surroundings that matter form the *context*: obstacles, which the robot must clear, and drivers, which move the target. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at τ_t.

3.2. Metric Scene and Motion Inputs

FoveaWAM encodes the target's motion precisely and its context through the current geometry: the current RGB-D frame gives the metric layout of the scene, and a short history of target poses gives the motion that one frame does not show, which also reflects how drivers have moved the target.

Current scene. FoveaWAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the forecasts and the actions. A frozen MAE ViT-B/16 [16] turns the RGB image into 196 patch tokens on a 14×14 grid, and the depth map back-projects the center of patch i to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

Target motion. Instead of past frames, from which motion must be recovered by comparing images, FoveaWAM receives the target's recent motion as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units. The velocity and acceleration that decide where the target will be thus enter the model directly, at the cost of a few low-dimensional tokens; Sec. 4.3 removes this history to measure its effect. In simulation, we use ground-truth target poses from the simulator. On the robot, a causal tracker estimates them from the RGB-D stream.

World stream. The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream.

3.3. Role-Aware 4D Predictive Supervision

FoveaWAM supervises its representation with two geometric forecasts of different precision: the motion of the whole scene as a sparse 3D flow on the patch grid, and, because the target's pose decides contact, the full SE(3) trajectory of the target. Both are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50$ ms for $h = 1, \dots, 8$, from 50 ms to 400 ms after capture; together with the current state, these horizons cover the execution times of all commands in the chunk.

Scene flow. At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for. For each of the 196 points $\mathbf{x}_i$ and each horizon h, a linear head on the final state of patch token i predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and

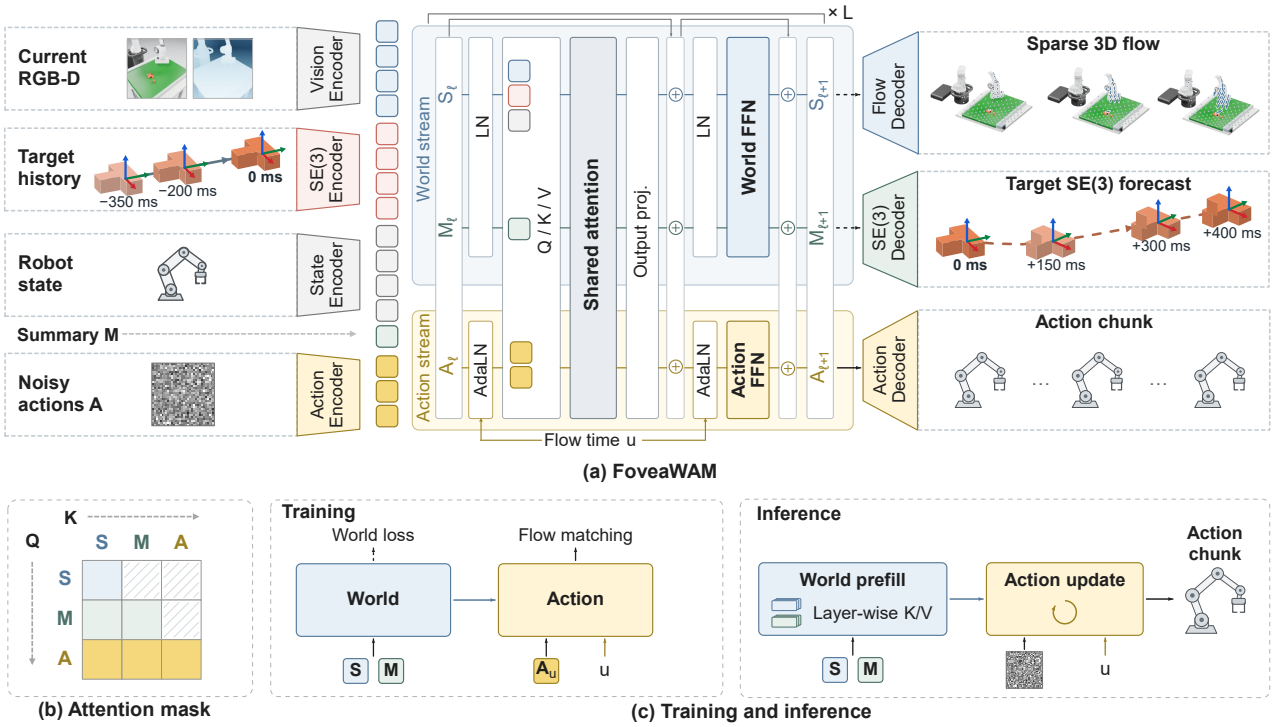


Figure 3. **FoveaWAM architecture.** (a) Current RGB-D and target/robot histories form scene tokens S, alongside learned summaries M and noisy actions A. Each layer shares masked attention, applies one World FFN to S, M and a separate Action FFN to A, and retains both residual updates. Final patch and mean-summary states support training-only geometric readouts (dashed); actions read hidden states. (b) Mask rows are queries and columns are keys. (c) Joint training combines auxiliary supervision and action flow matching. At deployment, world prefill caches layer-wise keys and values for action sampling, skipping geometric readouts. The RGB thumbnail is a Blender rendering. The target history (solid) and future SE(3) trajectory (dashed) meet at the current pose (0 ms). The future trajectory and sparse flow illustrate supervision from the recorded rollout (flow enlarged for display). Token counts, layer stacking and action icons are schematic.

a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where F_{ih} is the displacement of the surface point at $\mathbf{x}_i$, o_{ih} its visibility, s a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

Target trajectory. The flow samples the target only at the few patches it covers, too sparse to determine its rotation, so a dedicated linear head on the mean of the final summary states forecasts its rigid motion directly: for each horizon h, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\phi}_h$, trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h\in\mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{3}$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t+\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t+\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\phi}_h^{\wedge})R(\tau_t)$, $d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. The rotation term matters for contact: a rotation error θ can displace a contact point at distance r from the target's origin by up to $2r\sin(\theta/2)$. For example, a 5° error at $r = 10$ cm can shift the contact point by nearly 9 mm.

3.4. World–Action Learning and Inference

Action stream. The action stream generates the chunk by conditional flow matching [26]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command. In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With S, M, and A denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$,

and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in X attend to tokens in $\mathcal{Y}$. The world stream never reads the actions. The action stream reads, in every layer, both the patch states shaped by scene-flow supervision and the summary states shaped by target-trajectory supervision.

Training objective. For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \tag{4}$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \tag{5}$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\text{aux}}$ sums auxiliary appearance losses.

Inference. At deployment, FoveaWAM skips the forecast heads, as recent world–action models do [24, 54, 57]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary. On the robot, the latency of a query also includes tracking the target in the newest frame. Further implementation details are in the supplementary material.

4. Experiments

We test whether role-aware precision and low latency together improve manipulation when the scene keeps moving during inference. We first compare FoveaWAM with existing policies, each charged with its own latency (Sec. 4.2), then analyze which parts of its design account for the results (Sec. 4.3), and finally verify it on a real robot with tracked target poses (Sec. 4.4).

4.1. Experimental Setup

OnTimeBench. Unlike the evaluators of DOMINO [14] and PhysMani-Bench [55], which advance the simulation only after a policy has returned its prediction, OnTimeBench shares one physical clock among observation, scene motion, and action execution, so the scene keeps moving while a policy computes. It contains fourteen tasks in five families, built in MuJoCo/robosuite [43, 63]: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 4). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all OnTimeBench results on 1,400 held-out scenes (100 per task).

Evaluation. During each query, physics advances by the method's inference latency, drawn from its latency distribution measured in its native deployment setting at batch size 1 on one RTX 5090, the GPU of our robot. All methods share one executor, which executes each chunk from the command meant for the control-period boundary at which the chunk arrives. Besides this native setting, we add 150 ms of latency to every method and, as a reference without the cost of computation, pause physics during inference. We report the task-balanced success rate (SR) and the p95 time-to-action, the measured time from an observation to its first command; paired 95% intervals come from a bootstrap over held-out scenes. The supplementary material gives the full evaluation protocol.

Baselines. We compare with the reactive policies ACT [60] and Diffusion Policy [10] and the VLA $\pi_{0.5}$ [5]; the dynamic VLAs DynamicVLA [48] and PUMA [14]; the WAM OpenWAM-α [45]; and the dynamic WAM DynamicWAM [29] (Table 1). Each baseline is trained on the same demonstrations with its official architecture and a task-adapted recipe. FoveaWAM receives a single agent-view RGB-D frame and a history of ground-truth SE(3) target poses from the simulator. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Implementation. The transformer of FoveaWAM has 12 layers of width 1024 with 16 heads and 271.8M trainable parameters on top of the frozen MAE ViT-B/16 encoder. On OnTimeBench, it trains from random initialization with fused AdamW for up to 20 epochs. We evaluate the final checkpoint of the seed-0 run after 10 epochs (25,870 updates). Evaluation uses compiled BF16 and five Euler steps with TF32 disabled. Compilation and recorded-observation warm-up finish before physics starts. We then reset the history and inference seed.

4.2. Dynamic Manipulation Results

Comparison with baselines. With every method charged its own latency, FoveaWAM reaches 82.1% SR on held-out OnTimeBench scenes (Table 1), 20.7 points above the strongest baseline, PUMA (61.4%; 95% paired interval [18.4,

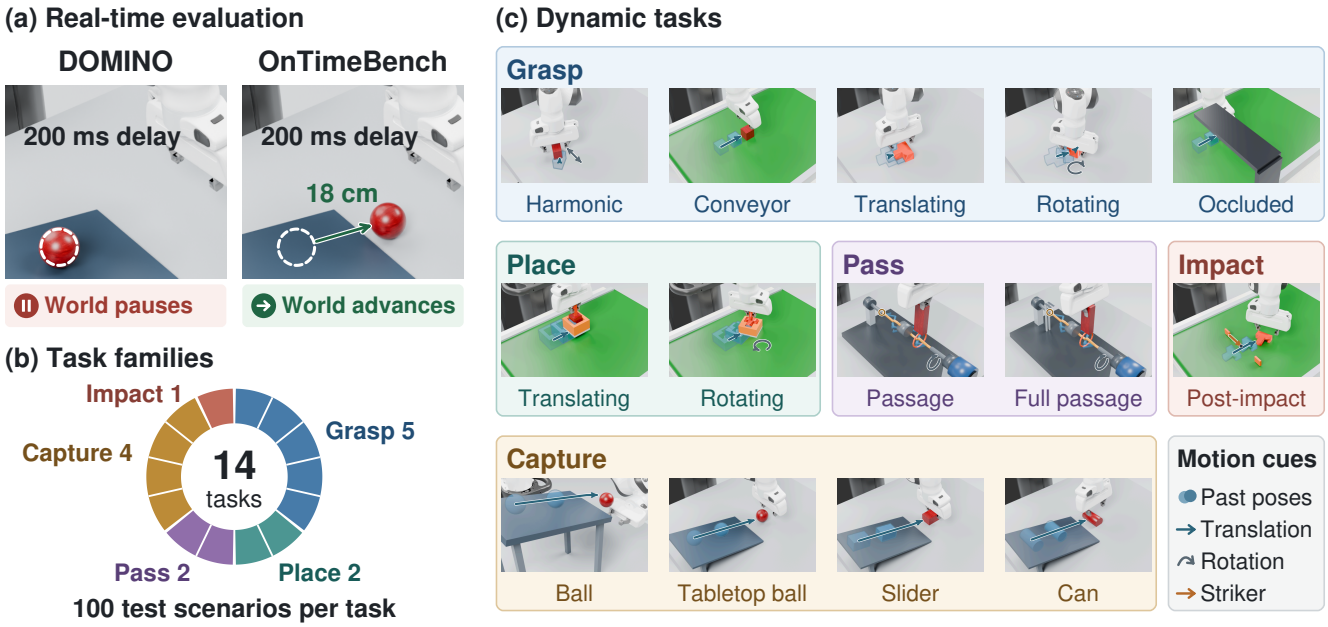


Figure 4. **OnTimeBench overview. (a)** The ramp example from the introduction illustrates a 200 ms inference delay. DOMINO's PUMA evaluator [14] pauses physics during inference. In real time, physics advances and the ball moves 18 cm from its observed position (dashed circle). **(b)** Fourteen tasks in five families with 1,400 held-out test scenarios. **(c)** Expert rollouts rendered with past poses and motion cues.

Table 1. **Results on OnTimeBench.** Task-balanced SR (%) on 1,400 held-out scenes with each method's native latency, with 150 ms added, and with physics paused during inference. p95: time-to-action in each method's native deployment setting on one RTX 5090. History: inputs that carry past motion beyond the current frame. OpenWAM-α uses its official caching and compilation, and (off) disables both. PUMA is timed with its official three-camera input; the other baselines with two cameras; FoveaWAM runs compiled BF16 inference on one RGB-D view and ground-truth target poses from the simulator. Best and second best per column are highlighted.

Method	History: Wrist RGB	History: Flow	History: SE(3)	SR (%) ↑: Native	SR (%) ↑: +150 ms	SR (%) ↑: Paused	p95 (ms) ↓
ACT [60]				41.6	24.6	40.6	**5.8**
DP [10]	✓			31.9	16.1	32.0	68.9
$\pi_{0.5}$ [5]				54.4	28.1	66.6	59.6
$\pi_{0.5}$+SE(3)			✓	58.7	28.0	76.0	61.2
DynamicVLA [48]	✓			51.1	23.5	54.9	97.3
PUMA [14]		✓		61.4	31.1	53.1	60.6
OpenWAM-α [45]				46.1	24.8	58.7	151.7
OpenWAM-α (off)				15.8	10.4	58.4	466.3
DynamicWAM [29]		✓		41.1	20.6	49.6	149.6
DynamicWAM+SE(3)		✓	✓	41.7	24.9	64.0	153.3
FoveaWAM			✓	**82.1**	**34.1**	**90.3**	15.1

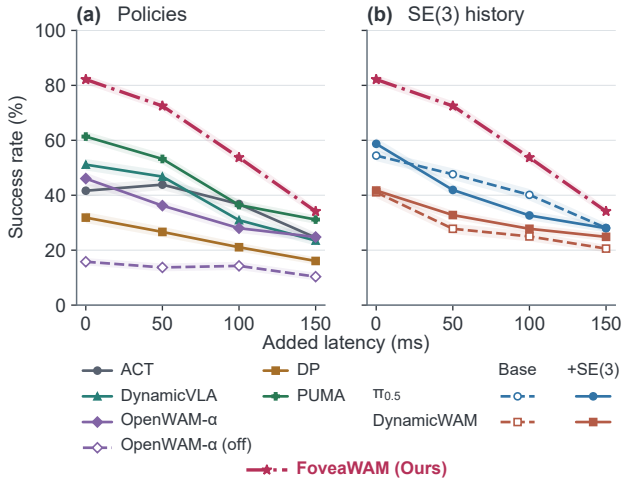


Figure 5. **Success under added latency.** Task-balanced SR on 1,400 held-out OnTimeBench scenes. Delay is added to each method's native latency, up to 150 ms, the largest step at which every method's chunk still covers the interval to its next chunk (Table S5 adds +200 ms). (a) Baseline deployments; off disables OpenWAM-α caching and compilation. (b) Base versus +SE(3)-history variants. PUMA is timed with its official three-camera input. FoveaWAM uses compiled BF16 (p95 15.1 ms). Bands: 95% task-stratified scene-bootstrap intervals.

23.1]). Target-motion input and low latency are both needed for this result. Given the same SE(3) history, $\pi_{0.5}$ and

DynamicWAM gain 4.3 and 0.6 points and reach 58.7% and 41.7%; ACT, the only faster method, reaches 41.6%. FoveaWAM and ACT are the only methods whose queries finish within one 50 ms control period, and every other baseline hands over its chunk at least two periods after capture. FoveaWAM has the highest or tied-highest SR on 13 of the 14 tasks (Table S4). Its margins over the best baseline are largest where a moving receptacle or the target's orientation decides success: 30 and 55 points on placement into translating and rotating receptacles, and 28 points on full wire passage.

Effect of inference delay. Latency alone can decide success. The two OpenWAM-α deployments run the same policy and agree with physics paused (58.7% and 58.4%), yet at native latency the one with a p95 of 151.7 ms reaches 46.1% and the one with 466.3 ms reaches 15.8%. Latency also erodes the value of motion input: the SE(3) history raises $\pi_{0.5}$ and DynamicWAM by 9.4 and 14.4 points with physics paused but by 4.3 and 0.6 at native latency. Charging latency therefore changes the ranking: $\pi_{0.5}$+SE(3), the strongest baseline with physics paused (76.0%), loses 17.3 points and falls behind PUMA. FoveaWAM leads in both settings, by 14.3 points over $\pi_{0.5}$+SE(3) and 20.7 over PUMA. Under added latency (Fig. 5), FoveaWAM remains the best method at every delay, ahead of the strongest baseline by 19.3 points at +50 ms, 13.6 at +100 ms, and 3.0 at +150 ms.

Results on DOMINO. To test the design beyond OnTimeBench, we train FoveaWAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [14] with its demonstrations and evaluate them with its native evaluator. As on OnTimeBench, FoveaWAM reads ground-truth target poses from the simulator. This evaluator pauses physics during inference, so FoveaWAM gains nothing from its low latency here; it still reaches the highest overall SR, 46.8%, 19.9 points above the strongest baseline, DynamicWAM (26.9%), and the highest manipulation score (Table 2). It leads on the 16 Track tasks, 55.5% against 31.9% for DynamicWAM; PUMA is strongest on the 8 Intercept tasks at 36.3%. More results are in the supplementary material.

4.3. Analysis of Role-Aware Modeling

Six controls share the data, recipe, and executor of FoveaWAM (Table 3). The history control changes the input. The other five change only the supervision of prediction heads that are skipped at deployment, so they run the same inference at the same latency as FoveaWAM, and any difference comes from what the model learned to forecast. We analyze SR at native latency, the condition FoveaWAM is designed for, overall and on the predefined rotation-sensitive and context-active task groups.

Table 2. **DOMINO results** on 24 single-arm Franka clean-L1 tasks, with the same 2,400 test scenes per method (100 per task). Intercept/Track contain 8/16 tasks; All weights all 24 tasks equally. SR and MS use a 0–100 scale and retain failures. Best and second best are highlighted.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	23.8	8.1	13.3	27.8
DynamicWAM	16.8	31.9	26.9	41.4
PUMA	**36.3**	12.2	20.2	35.4
FoveaWAM	29.4	**55.5**	**46.8**	**62.0**

Table 3. **Analysis of role-aware modeling.** Task-balanced SR (%) of FoveaWAM on held-out OnTimeBench scenes and its change for each control (Δ, points). Red marks drops and green marks gains, with shading scaled within each column. Rot./Ctx.: rotation-sensitive and context-active tasks (Table S2). Table S7 adds paused physics.

	Success rate, all tasks		Groups (Native)	
	Native	+150 ms	Rot.	Ctx.
FoveaWAM	**82.1**	**34.1**	**80.4**	**82.7**
Precision allocation				
Uniform coarse flow	−4.5	−0.5	−8.6	−4.6
Uniform fine flow	+0.8	−0.9	−3.4	−2.6
w/o flow	−4.2	−1.4	−11.7	−3.9
Target motion history				
w/o history	−19.6	+0.2	−24.0	−2.3
Future supervision				
w/o future	−4.3	−3.4	−12.6	−2.1
w/o $\mathcal{L}_{\text{aux}}$	−2.6	−1.9	−11.6	+0.1

Table 4. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95: end-to-end time-to-action (ms) on one RTX 5090, including tracking for FoveaWAM. Best per column is highlighted, including ties.

Method	Belt grasp	Box place	Turntable	Toy car grasp	Tennis ball	Ramp grasp	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
ACT	—	—	—	—	—	—	—	—
FoveaWAM	—	—	—	—	—	—	37.5	—

Allocation of precision. Role-aware supervision gives the target a precision that uniform supervision does not. At the same flow density, replacing the SE(3) trajectory with uniform flow reduces SR by 4.5 points overall and by 8.6 on rotation-sensitive tasks. Raising the uniform flow to a 56 × 56 grid, 16× the points, costs [X.X]× the training time of FoveaWAM. It recovers the overall SR (+0.8) but still trails the role-aware model on both task groups, by 3.4 points on rotation-sensitive tasks and 2.6 on context-active tasks. One target trajectory and 196 flow points thus

serve these tasks better than 3,136 uniform flow points, at a fraction of the training cost. The sparse flow is needed as well: removing it while keeping the SE(3) trajectory reduces SR by 4.2 points overall, by 3.9 on context-active tasks, and by 11.7 on rotation-sensitive tasks. The two forecasts are complementary, and FoveaWAM has the highest SR on rotation-sensitive tasks among all variants. All three controls keep the visual auxiliary objectives.

Target motion history. The target-motion history is the largest single factor. Keeping the current target pose but dropping its history, including the velocity and acceleration features, reduces SR by 19.6 points overall and by 24.0 on rotation-sensitive tasks. A single pose does not show how the target moves; the metric history supplies the motion that the commands must anticipate.

Future supervision and auxiliary objectives. Supervising the future, rather than only reconstructing the current target, is worth 4.3 points overall and 12.6 on rotation-sensitive tasks, and 3.4 points with 150 ms added, although the forecasting heads never run at deployment. The visual auxiliary objectives $\mathcal{L}_{\text{aux}}$ add 2.6 points overall and 11.6 on rotation-sensitive tasks. Every component of the role-aware design therefore contributes: removing any one of them lowers SR by 2.6 to 19.6 points overall and by 8.6 to 24.0 on rotation-sensitive tasks, where contact depends on the target's orientation.

4.4. Real-Robot Evaluation

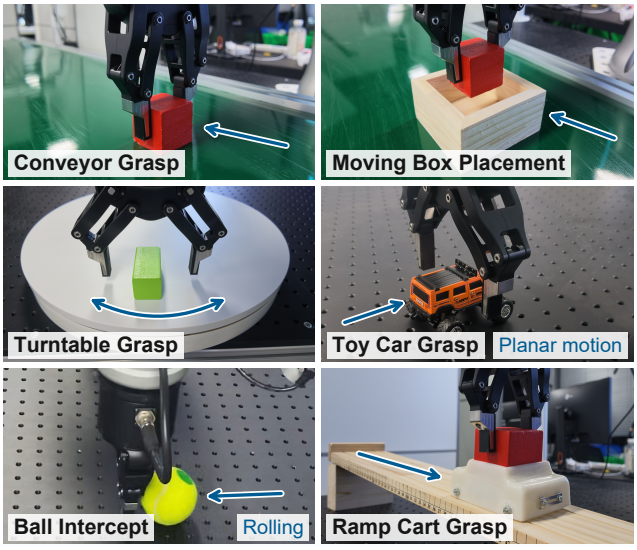


Figure 6. **Dynamic tasks on the real robot.** Straight arrows show example pre-contact motion of the target or moving box. The double-headed arc denotes turntable rotation. The tennis ball image is a demonstration frame.

Setup. The six tasks on a Flexiv Rizon 4s arm with a GN01 gripper include rolling tennis-ball interception (Fig. 6 and Table S8). Each method draws training data from a shared corpus of 600 teleoperated demonstrations. We plan 20 trials per task and method. FoveaWAM starts from a DROID-pretrained base [21]. In the planned evaluation, it uses RGB-D and online target pose estimates from the fixed camera only. The baselines ($\pi_{0.5}$, ACT, and DynamicWAM) use RGB from the fixed and wrist cameras.

Results. FoveaWAM succeeds in 37.5% of all trials, against [XX.X]% for [the strongest baseline] (Table 4). Its p95 latency, including tracking, is [XX] ms, [within one control period], whereas [name] needs [XX] ms. The margin is largest on [tasks]. [Whether the ranking of Table 1 carries over to real perception, and what tracking costs.] [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of FoveaWAM or a failure of a baseline.] The dominant failure of FoveaWAM is [failure mode]. Additional details are in the supplementary material.

5. Conclusion

FoveaWAM spends precision where contact depends on it: a metric SE(3) history and forecast for the target and a sparse 3D flow for the whole scene, with the forecasts used only as training supervision. When every method is charged its own latency, these choices let FoveaWAM act both precisely and quickly: it outperforms reactive policies, VLAs, and world–action models, including ones built for moving targets, on OnTimeBench and on six real-robot tasks at a p95 latency of 15.1 ms on OnTimeBench with compiled BF16 inference, and it also leads on DOMINO. The strongest baseline is PUMA under native latency and $\pi_{0.5}$+SE(3) in the paused column. At native latency on OnTimeBench, removing target pose history and removing future supervision reduce success by 19.6 and 4.3 percentage points, respectively. We will release OnTimeBench so that future dynamic policies are evaluated with their latency charged.

Limitations. Deployment requires a specified rigid target and reliable online pose estimates. The target forecast predicts one future per horizon and ignores the effects of robot contact on target motion. Uncertainty-aware forecasts and contact-conditioned dynamics could address these limits. Independently moving context lies outside our task set and would need its own motion history, for example from scene flow.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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